

Layout and precision changes beyond the $a_b/2$ ceiling

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Conditional status. Every statement here assumes the pinned upstream interfaces and these unmerged results on `CrocSwap/integer-mult-bounds`: PR #5's chirped-correlation Gaussian lemma and its choice of α ; PR #3's compressed complex network; and PR #7's networks, with $a_b = 3/(4 \cdot 10^8)$, $a_c = 39/10^9$, $m = 21952$ and $s = 45772350635112192$. These are the round-two values; the round-three savings, and the further results they assume, are in `round3-combination.tex`; the round-four savings are in `data-edge-batching.tex`, `complex-source-frames.tex` and `scripts/certificate_round4.py`.

Deliberate departures from upstream hypotheses. The prime-interval condition $1 - 2\epsilon > 0$ is replaced by the prime number theorem (Section 1). The resampling hypothesis $\theta > p/\alpha^4$ is replaced by $\alpha^2\theta \geq 1$, and the over-sampling is reduced to $\theta \approx 1/(4dL^y)$ (`segmented-inverse.tex`). The band $r \geq 2^{a_r p/d}$ is replaced by $r \geq 2^{p^\mu}$ (Section 1). $K = o(\ell)$ is dropped (Section 2). The guard exponent uses the certified bound $\log_m s < 3.8376$ in place of 5; the witness also passes with 5.

Throughout, $L = \log_2 T$ is the address length, p is the coefficient precision, $d = \Theta(L^\epsilon)$ is the number of axes and $\ell = \Theta(L^{1-\epsilon})$ is the axis width. Costs are divided by the logical volume $V = \Theta(Tp)$ and expressed as powers of L . Section and lemma names refer to the pinned upstream source (`upstream/build/sections`), to Colkitt's compact-control notes (`notes/compact-control-*.tex`), and to PR #5 (`notes/fast-gaussian-resampling.tex`).

1 Idea D: coefficients longer than $\log n$

Scope. Idea D removes only the precision part of the $\epsilon < 1/2$ wall. On its own it does not move κ . PR #5's Gaussian row, $d^2 p^\delta$, comes from

about $1/\theta \approx 4d$ Neumann terms per axis, and it still caps $\epsilon < 1/2$ whatever the precision. Going past $\epsilon = 1/2$ needs Idea D together with Idea B' (`segmented-inverse.tex`). PR #5's chirped correlation is required: upstream's direct Gaussian windows cost $\sqrt{d}p^{1+\delta}$ per bit once $p \gg L$.

Change. Upstream splits each input into $b = \lceil \log_2 n \rceil$ -bit digits and sets $p = 6b$. Instead use $b' = \lceil b^{1+x} \rceil$ -bit digits for a fixed rational $x > 0$, still with $p = 6b'$. Then $T \in [4n/b', 8n/b']$, so $Tp = \Theta(n)$, and $L = b - (1+x)\log_2 b + O(1)$. Set $d = \lfloor b^\epsilon \rfloor$ and choose r, ℓ from T as upstream does.

Consequences to check.

1. *Capacity and rounding* (`08-assembly.tex`, eq. capacity and eq. final-error). The coefficients satisfy $c_j \leq q(2^{b'} - 1)^2 < 2^{3b'}$, since $q \leq 2^{b'}$. The final error is $2^{2b'+4}S \cdot 28 \cdot 2^\gamma T^2 L \cdot 2^{-6b'} < 448 b' 2^{-b'+\gamma}$, using $S \leq T < 2^{b'}$ and $L < b'$. This is below $1/2$ whenever $\gamma \leq b'/4$.
2. *Gaussian width.* Take $\alpha^2 = \lfloor Q/(48d) \rfloor$ with $Q = p = 6b'$, that is $\alpha^2 = \lfloor b'/(8d) \rfloor$, and oversampling $\theta_i \approx \eta = 1/(4dL^y)$ (`segmented-inverse.tex`, last section). Then $\gamma = 2d\alpha^2 \leq b'/4$ and $\alpha^2\theta_i \geq b'/(32d^2L^y) \geq 1$. The second holds eventually when $2\epsilon + y < 1 + x$.
3. *Record regime.* Every use of $r \geq 2^{a_r p/d}$ (`05-layers.tex` lines 20–32 and 948–962, `06-transforms.tex` 297 and 384, `08-assembly.tex` 96, 168 and 293) only needs r to outgrow every fixed power of p . Here $r = 2^{\Theta(L/d)} = 2^{\Theta(b^{1-\epsilon})}$ and $p = \Theta(b^{1+x})$, so that still holds. Replace the hypothesis by $r \geq 2^{p^\mu}$ for a fixed $\mu > 0$.
4. *Address bound.* $\log_2(2M) \leq C_M p$ only becomes easier.
5. *Chunk-exchange row.* Upstream writes it as $pK^{\tau-1}$ by counting $O(p/K)$ exchanges from $d\ell = O(p)$. The true count is $d\ell/K = L/K$, so the row is $LK^{\tau-1}$. Idea C makes it zero.
6. *Guard.* The coefficient-depth bound $\Delta = C_0 d^{C_1}$ must be $O(p)$. That needs $\epsilon C_1 < 1 + x$, measured in powers of b .
7. *Prime selection* (`08-assembly.tex` 120–160). The cited explicit interval lemma needs $\log x > 8d$, which fails for $\epsilon \geq 1/2$. Replace it with the prime number theorem. Each interval has length $\eta x = x/(4dL^y) = x/(\log x)^{O(1)}$, so the classical prime number theorem with error term $O(x \exp(-c\sqrt{\log x}))$ already gives more than d primes in each interval

for every fixed $\epsilon < 1$, with effective constants. The machine only needs existence; it scans and tests by trial division in $n^{o(1)}$ time, since the log of the scan length is $O(\log p + L/d) = o(L)$.

8. *Per-bit costs.* Every row whose cost grows with precision carries only a factor $p^\delta = b^{(1+x)\delta}$. The pointwise products cost $\log(rp) = \Theta(\ell + \log p)$ per bit.

2 Idea C: one chunk per axis

Change. Upstream takes $K = \lfloor d^c \rfloor$ because the layer cost had a factor K^τ . Colkitt’s compact controls removed that factor. Set $K = \ell - 1$ instead. In the transform-layout lemma (`06-transforms.tex`, `lem:synthetic-transform-cost`), this gives $b = (\ell - 1) \bmod K = 0$ and $q = 1$. The target chunk order then equals the source order, so the procedure performs no chunk exchanges. The prefix holds only the top slot of each long axis.

Cost. The layout costs $O(Vd)$ and the rounds cost $O(V\ell d^{\lambda'})$. The rows dK and $pK^{\tau-1}$ become d and 0.

Layer hypotheses to recheck (`notes/compact-control-layout.tex`).

- $K/\log p \rightarrow \infty$ and $K \geq G + 4\ell_p + 10$ hold.
- The reservation count is $q_F + q_B \leq 3dG/K + 2 = O(d^2 \log p/L) + 2$. In powers of d this has exponent $(2\epsilon - 1)/\epsilon$, which must stay below λ' , that is $\epsilon < 1/(2 - \lambda')$. Upstream’s guard implied this, but Idea D’s relaxed guard does not, so the certificate checks it separately.
- The simultaneous-layer proposition fixes $K = \lfloor d^c \rfloor$ (`05-layers.tex` lines 21, 852 and 954–958). Restate it with K supplied as input, in the band $\frac{1}{2}d^{(1-\epsilon)/\epsilon} \leq K \leq p$. Nothing else uses $K = o(\ell)$: its only uses are $K \leq \ell - 1$ and the row dK .
- The bound $K \geq d^c/2$ used in that note is replaced by $K \geq \frac{1}{2}d^{(1-\epsilon)/\epsilon}$.
- The row index uses $q_0 = O(\log d)$ whole axes, unchanged.

3 Idea A: expose only the fine bits of an axis

The full statement and proof, with a 12-tape schedule, the corrected input window and the movement bounds, are in `notes/fine-field-lemma.tex`. `scripts/check_fine_field.py` simulates every data movement on explicit tapes and checks outputs and movement bounds. The summary below is superseded where it differs. In particular, a block reads input slabs $J', J' + 1, J' + 2$ with $J' = \lfloor (c(J2^f) - W)/2^f \rfloor$, the lowest input read, not $J' - 1, J', J' + 1$.

Change. The tensor interface (`07-resampling.tex`, `lem:tensor-resampling`) makes each axis the innermost address field before running its line algorithm, at cost $O(Tpd^2(1 + \ell^\tau))$ upstream and $O(Tpd(1 + \ell^\tau))$ with nonadjacent routing. Instead, write the axis- i coordinate as $j = J2^f + u$ with $f = \lceil \log_2(cW_{\text{line}}) \rceil$. Here W_{line} is the largest window the line algorithm reads: $3m + 3$ for S' , a block length λ plus $2w$ for the inverse of Idea B', and the periodic halo. Move only the f -bit field u to the innermost address position, leaving J in place.

Execution. The physical order becomes (X, J, Y, u) , where X and Y are the fields of other axes before and after J . For fixed X , the records with a given J form a contiguous slab of $|Y|2^f$ records.

- Every block of the line algorithm reads inputs from at most three consecutive slabs, $J' - 1, J', J' + 1$. The input slab index J' is a non-decreasing function of the output slab index J , roughly $J' = \lfloor J/\sigma \rfloor$ for expansion, and the identity for N .
- Keep three read heads on copies of the input array, offset by one slab, and advance them in step with the output over Y and J . When J' repeats, the heads rewind at most three slabs per output slab, so the total movement is still $O(V)$. These are three tapes, each holding a copy; copying costs $O(V)$ per pass.
- The period s_i is not a multiple of 2^f . The wrap partner of slab 0 is the slab containing $s_i - 1$, and the window must be spliced at $u = s_i \bmod 2^f$. For each X -group, copy the boundary slabs (about $3|Y|2^f$ records) to a scratch tape and splice from it, at $O(V)$ per pass. The same applies to every application of E , whose period is also s_i .
- *Per-chunk setup.* There are $T/2^f$ chunks and $2^f = \Theta(W_{\text{line}})$, so upstream's argument that per-line setup is negligible does not apply

directly. Compute the block offsets and q -counters once per J and copy $O(p)$ bits per chunk. Run the validity test from an odometer over Y with per-axis flags, at $O(\ell)$ per chunk. Allow one extra partial block per chunk, at $O(mp^{1+\delta})$. Each chunk has volume at least $cW_{\text{line}}p \gg \ell^2, p$, so all of this is $O(\text{chunk volume} \cdot p^\delta)$.

- The backward sweep of Idea B's cyclic block-tridiagonal solve carries state along the line. It uses one more tape for the previous output slab at the same Y , and reads fixed-width records right to left.

Cost. Exposing and restoring the fine field are two interchanges of f -bit fields, $O(Vf^\tau)$, or $O(Vf)$ by single-bit moves. Over all axes this gives $O(Vd \log L)$, with exponent $\epsilon + o(1)$.

4 Idea E': reversing the CRT axis order by permuted selected-bit additions

Problem. The Chinese remainder layout (`08-assembly.tex`, explicit cyclic-convolution layout) reverses the order of the d axes so that every target follows its controls. With chunk interchanges this costs $O(Vd\ell^\tau)$, which is the row $(1 - \tau)(1 - \epsilon)$ and caps κ at $a_b/2$.

Primitive. Colkitt's Proposition 1 (`artifacts/compact-control-note.pdf` in his repository, compact-control selected-bit addition) toggles, for slots x, y of fK bits with selected positions $j_i = \rho + iK$, each y_{j_i} by x_{j_i} at cost $O(V((f \log p)^\tau + 1))$.

- Every rotation offset in that construction is computed from fields that precede the rotated field, by Lemma 1 of the note.
- Replacing the control $z_i = x_{j_i}$ of target segment i by $x_{j_{\pi^{-1}(i)}}$, for a fixed permutation π , changes only those offset formulas:

$$\sum_i 2x_{j_{\pi^{-1}(i)}} t_i 2^{j_i}, \quad \sum_i x_{j_{\pi^{-1}(i)}} (1 - 2t_i) 2^{j_i}, \quad \sum_i (y_{j_i} \oplus x_{j_{\pi^{-1}(i)}}) B^i.$$

All three are still computable from preceding fields in polynomial time.

- The guards, the dirty temporaries u, t, b , and the repair argument do not depend on the pairing. The bad-address predicate is a function of the same digits.

So the permuted selected-bit addition $y_{j_{\pi(i)}} \oplus = x_{j_i}$ has the same cost. The four-update identity holds for any control bit that stays fixed during it, and x is never written. The bad-address predicate reads only the t, u digits and the guard bits g_i , which contain no selected bit, so the ideal map still preserves the bad set.

Reversal. Let x be axes $1, \dots, d/2$ and y axes $d/2 + 1, \dots, d$, each a slot of $f = d/2$ chunks with $K = \ell - 1$, after the top bits of the long axes are extracted (see the placement constraints below). Let π reverse the chunk order. For each level $\rho \in [0, \ell)$, three permuted additions $x \oplus = \pi^{-1}y$, $y \oplus = \pi x$, $x \oplus = \pi^{-1}y$ exchange bit ρ of axis i with bit ρ of axis $d + 1 - i$ for every $i \leq d/2$. The temporaries are reserved from existing axes as in Colkitt's layout note. The $O(d^2 \log p/L)$ reserved axes are reversed by single-bit moves at $O(V\ell)$ each, so $O(Vd \log p)$ in total. Unequal widths $\ell, \ell - 1$ cost $O(d)$ single-bit moves.

Placement constraints. These fixes are needed.

- *Record regime.* Lemma 1 and the repair need record width beyond every power of p . At the CRT stage the records are single coefficients. Treat the innermost axis as the record suffix, keep it out of both slots, and move it by ℓ single-bit moves at $O(V\ell)$. The back field b sits between y and that suffix.
- *Equal-width chunks.* First extract the top bits of the long axes by $O(d)$ single-bit moves, then use $K = \ell - 1$. Alternatively, choose a long/short pattern that is symmetric under reversal.
- *Temporaries.* The fields u, t (about dG bits) before x and b after y come from front and back axes excluded from the swap. Reverse the middle axes in place, then re-place the reserved axes and the record axis by single-bit moves, at $O(Vd \log p + V\ell)$.

The certificate uses the better of this row and the pairwise-swap row $(1 - \tau)(1 - \epsilon)$.

Cost. The reversal costs $O(V\ell(d \log p)^\tau)$, an exponent of $1 - \epsilon(1 - \tau) + o(1)$, so the margin is $\epsilon(1 - \tau)$. This exceeds $\kappa < \epsilon(1 - \lambda')$ because $\lambda' > \tau$.

5 Resulting margins

See `certificate.py`. The rows are:

- prefix moves, $1 - \epsilon$;
- butterflies, $\epsilon(1 - \lambda')$, which binds;
- fine exposures, $1 - \epsilon - o(1)$;
- the Gaussian maps with recentred sub-blocks (`segmented-inverse.tex`, last section), $1 - \epsilon - o(1)$;
- chirps, $1 - \epsilon - o(1)$;
- pointwise products, ϵ ;
- reserved axes, $1 - \epsilon - o(1)$;
- CRT reversal, $\epsilon(1 - \tau)$.

The side conditions are the guard $\epsilon C_1 < 1 + x$; precision $2\epsilon + y < 1 + x$, where the oversampling is $\theta \approx 1/(4dL^y)$; sub-block coupling $y \geq \epsilon$; and λ' above τ , σ , χ and the leaf exponent.

A Finite payload alphabets

The bit network of PR #7 carries ternary payloads. The pinned interchange lemmas are stated for bit arrays and pointwise bit gates (`04-swap.tex`: the circuit transfer before `prop:power-interchange`, its descriptor paragraph, and `lem:chunk-swap`). The witness needs their versions for a fixed finite alphabet.

Lemma 1 (Finite alphabet). *Let Σ be a fixed finite alphabet. Replace “bit array” by “ Σ -array” and “pointwise bit gates” by “pointwise gates on Σ ” in `lem:elementary-streams`, `prop:power-interchange` and `lem:chunk-swap`. The statements still hold, with the same exponent τ and constants that depend on $|\Sigma|$.*

Proof. Encode each symbol of Σ as a fixed-width record of $B_0 = \lceil \log_2 |\Sigma| \rceil$ bits. This adds one innermost suffix field $[2^{B_0}]$, which `lem:chunk-swap` allows ($[B]$ in its statement). Volumes change by the fixed factor B_0 .

- Every operation other than a gate is an address permutation. It moves whole records and never reads their contents, so it is unaffected.

- A pointwise gate on Σ is a fixed finite map applied to records at matching addresses of a fixed number of aligned role streams. That is item 3 of lem:elementary-streams with records of B_0 bits, so it costs $O(V)$.
- The frame identity in prop:power-interchange uses only that a common address permutation commutes with every pointwise gate. That holds for any alphabet.
- The descriptor paragraph assumes a one-bit payload only to bound $V_j \geq q^{2e_0}$. With B_0 -bit records the volume is larger by a factor B_0 , so the bound still holds.

The recursion, its rank count s , the role split and the depth-first schedule do not refer to payload values, so the recurrence $F_k \leq (s/W)F_{k-1} + O(1)$ and the bound $O(Ve^\tau)$ hold unchanged. $\square$